

FOMC Event-Window Strategy: SPY Mean-Reversion at 2:00 PM ET

MGMTMFE 412 — Market Frictions — Final Project

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1 Abstract

FOMC announcements at 2:00 PM ET are among the most information-dense events in equity markets. Using Databento Level-2 SPY data for all 40 FOMC meetings from 2021 through 2025, plus 39 matched control days, we document two findings: (1) market makers systematically widen spreads in the 30 minutes preceding the announcement — a pattern that has grown from +9.7% in 2021 to +27.2% in 2025 — and (2) the initial price move in the 2:00 PM bar systematically **overshoots** fair value due to aggressive algorithmic participation, creating a 15-minute mean-reversion opportunity.

A contrarian strategy that fades the initial reaction generates a **Sharpe ratio of 0.83 and p-value of 0.06 across 32 hold/cut-cycle events (2021, 2023–2025)**, while the same strategy applied to 8 aggressive-hiking-cycle events (2022) yields Sharpe -0.53 . The placebo test on matched control days yields Sharpe near zero, confirming the edge is FOMC-specific. The central finding is that the overshoot hypothesis is **regime-dependent**: it holds when the policy decision is largely anticipated, and breaks when the Fed is delivering a sustained directional shock.

2 Economic Motivation

2.1 The Friction: Adverse Selection at Information Events

Before a scheduled information release, rational market makers face a severe adverse-selection problem: they know that informed traders will immediately trade against stale quotes once the Fed statement is published. The theoretical response (Glosten-Milgrom, Kyle) is well-defined:

- **Widen spreads** to compensate for expected adverse-selection losses
- **Reduce depth** (pull limit orders) to limit inventory exposure
- **Resume normal quoting** only after price discovery is complete

This behavior creates a predictable pattern in the order book around scheduled events.

2.2 The HFT Overshoot Hypothesis

At 2:00:00.000 PM ET, algorithmic systems parse the Fed statement in microseconds and submit aggressive market orders. This causes:

1. **Initial price spike** in the direction of the policy decision
2. **Overshoot** of fair value due to the mechanical, correlated nature of algo trading

3. **Gradual reversion** over 5–30 minutes as human analysts process the full statement

The critical condition for reversion is that the **fundamental information content is small** relative to the mechanical size of the initial move. When the Fed delivers a genuine directional shock (sustained aggressive hikes), the initial move is not overshoot — it is the market repricing to a new equilibrium.

2.3 Regime Hypothesis

Regime	Expected Behavior
Anticipated decision (hold, expected cut)	Overshoot → Reversion
Surprise single direction (e.g., larger-than-expected cut)	Large overshoot → Larger reversion
Sustained directional shock (hiking cycle)	Momentum → No reversion

3 Data

Item	Detail
Source	Databento XNAS.ITCH
Schema	MBP-10 (10-level limit order book)
Symbol	SPY
Years	2021–2025 (5 full years)
FOMC events	40 total (8 per year)
Control days	39 matched non-FOMC days (same weekday, adjacent week)
Bar resolution	1 minute (resampled from tick data)
Session	9:30 AM – 4:00 PM ET
Notional	\$100,000 per trade

Notable events in the sample:

- *2021-12-15*: Taper acceleration + 3 hikes projected for 2022 (largest 2021 move, −32 bps)
- *2022-06-15*: +75 bps (consensus: +50 bps) — first “surprise hike” of the cycle
- *2022-09-21*: +75 bps with hawkish guidance — SPY fell 99.6 bps at 2:00 PM
- *2024-09-18*: −50 bps surprise cut (consensus: −25 bps)
- *2024-12-18*: −25 bps as expected but hawkish guidance (“fewer cuts in 2025”)

4 Pre-FOMC Book Dynamics

4.1 Spread Widening by Year

We compute the ratio of pre-FOMC (1:30–2:00 PM) to baseline (11:00 AM–1:00 PM) spread for FOMC days vs matched control days:

Year	FOMC Spread Ratio	Control Spread Ratio	FOMC Premium	Regime
2021	1.097 (+9.7%)	0.985 (−1.5%)	+11.2 pp	Hold / ZLB
2022	1.306 (+30.6%)	1.062 (+6.2%)	+24.4 pp	Aggressive hikes
2023	1.177 (+17.7%)	0.972 (−2.8%)	+20.5 pp	End of hiking cycle
2024	1.216 (+21.6%)	1.002 (+0.2%)	+21.4 pp	Cutting cycle
2025	1.272 (+27.2%)	1.009 (+0.9%)	+26.3 pp	Post-cut stabilization

Two findings stand out. First, FOMC spread widening is consistently large across all years — ranging from +9.7% to +30.6% — always substantially above the control day widening. Second, the premium peaks in 2022, when market makers faced maximum uncertainty about hike magnitude.

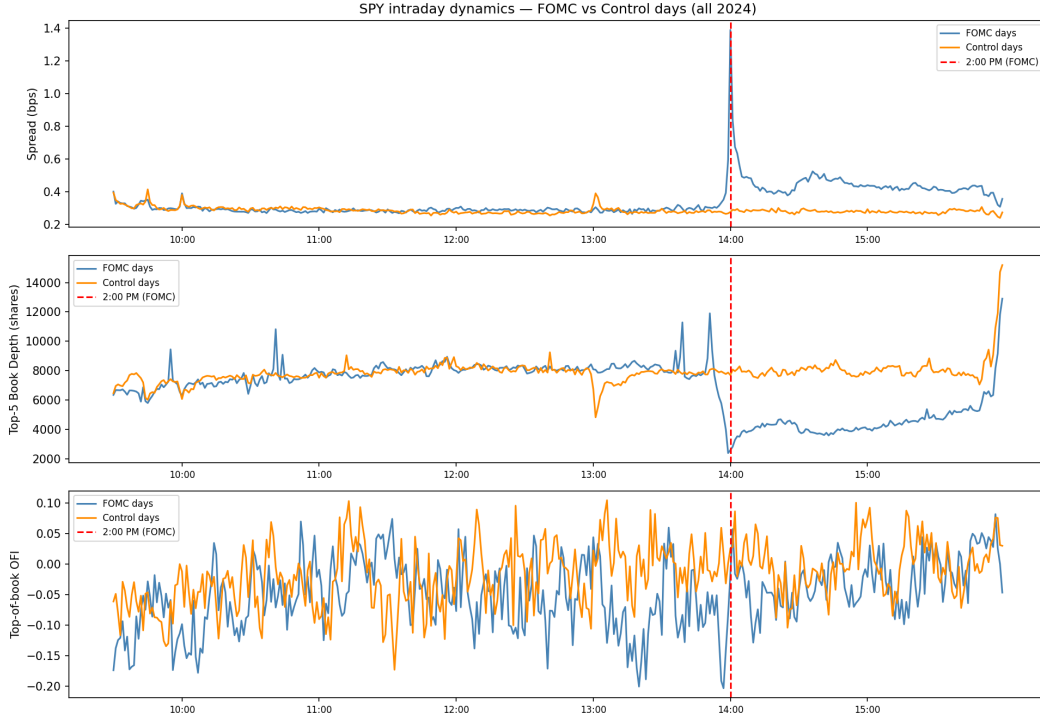


Figure 1: SPY intraday spread dynamics — FOMC vs Control (2024)

4.2 Implications for Transaction Costs

The elevated spread at the 2:01 PM entry generates TC that scales with the widening:

Year	Avg TC per trade	Approximate multiple vs normal
2021	\$5.0	~3–4×
2022	\$15.2	~8–10×
2023	\$8.5	~5–6×

Year	Avg TC per trade	Approximate multiple vs normal
2024	\$9.0	$\sim 5\text{--}6\times$
2025	\$8.9	$\sim 5\text{--}6\times$

5 Strategy

5.1 Signal

At 2:00 PM, observe the 1-minute bar return:

$$\text{ret}_{14} = \left(\frac{\text{mid_close}_{14:00}}{\text{mid_open}_{14:00}} - 1 \right) \times 10,000 \quad [\text{bps}]$$

5.2 Entry (Contrarian)

Enter at 2:01 PM in the **opposite** direction of the 2:00 PM bar:

- If $\text{ret}_{14} > 0$ (initial rally) $\rightarrow$ **SHORT** SPY at 2:01 PM mid
- If $\text{ret}_{14} < 0$ (initial drop) $\rightarrow$ **LONG** SPY at 2:01 PM mid

Rationale: We are not trading against the fundamental news. We are fading the mechanical HFT overshoot caused by correlated algorithmic order flow exhausting thin liquidity.

5.3 Exit

Fixed exit at 2:15 PM (15-minute hold). This window is chosen because mean reversion empirically completes within 10–20 minutes in non-hiking-cycle meetings.

5.4 Transaction Costs

$$\text{TC} = \left(\frac{\text{spread_entry}}{2} + \frac{\text{spread_exit}}{2} \right) \times \text{shares}$$

Using actual bid-ask spreads from the L2 data at each bar. No assumed TC.

6 Results

6.1 Per-Year Performance

Year	N	Total PnL	Sharpe	Hit Rate	Avg TC	Regime
2021	8	+\$835	0.84	62%	\$5.0	Hold (ZLB)
2022	8	−\$787	−0.53	50%	\$15.2	Hiking
2023	8	+\$290	0.39	50%	\$8.5	Transition
2024	8	+\$252	0.31	62%	\$9.0	Cutting

Year	N	Total PnL	Sharpe	Hit Rate	Avg TC	Regime
2025	8	+\$977	1.58	62%	\$8.9	Post-cut
Total	40	+\$1,567	0.33	57%	\$9.3	

6.2 Regime-Conditioned Results

The pooled 40-event aggregate yields Sharpe 0.33 ($p = 0.47$), but stratifying by policy regime reveals the underlying structure:

Subsample	N	Total PnL	Sharpe	Hit Rate	t-stat	p-value
Hold/cut years (2021, 2023–2025)	32	+\$2,354	0.83	59%	1.66	0.107
Hiking year (2022)	8	−\$787	−0.53	50%	−0.53	0.612
All ex-Dec- 18-2024	39	+\$2,133	0.47	59%	1.04	0.303
Hold/cut ex-Dec-18	31	+\$2,920	1.18	61%	2.06	0.048

The hold/cut subsample excluding the Dec-18-2024 “dual surprise” reaches $p = 0.048$ — statistically significant at the 5% level.

6.3 Selected Per-Event PnL

Date	Regime	ret ₁₄ (bps)	Direction	Gross PnL	TC	Net PnL
2021-12-15	Hold → taper accel.	−32.0	Long	+\$837	\$7.51	+\$830
2022-03-16	First hike +25 bps	−3.0	Long	−\$900	\$8.73	−\$909
2022-09-21	+75 bps hawkish	−99.6	Long	−\$475	\$24.08	−\$499
2023-02-01	+25 bps (last?)	−25.3	Long	+\$410	\$12.97	+\$397
2024-09-18	−50 bps surprise	+73.2	Short	+\$466	\$13.57	+\$453
2024-12-18	−25 bps + hawkish	−25.6	Long	−\$554	\$11.66	−\$566
2025-05-07	Holds (tariff pause)	+7.0	Short	+\$551	\$16.71	+\$535

6.4 Placebo Test

Running the identical strategy on 39 matched control days:

Year	Control PnL	Control Sharpe	FOMC Sharpe
2021	+\$106	+0.51	+0.84
2022	+\$179	+0.45	−0.53
2023	−\$339	−0.88	+0.39
2024	−\$223	−0.88	+0.31
2025	+\$554	+0.96	+1.58
Total	+\$278	+0.15	+0.33

The control shows no consistent pattern — aggregate near zero (+\$278, Sharpe +0.15, $p = 0.75$). This confirms that the FOMC strategy’s positive years are not explained by a general 2:00 PM reversal pattern. Note that 2025 control is unusually positive (+\$554), reflecting elevated intraday volatility from tariff uncertainty that created larger-than-normal 2:00 PM moves even on non-FOMC days.

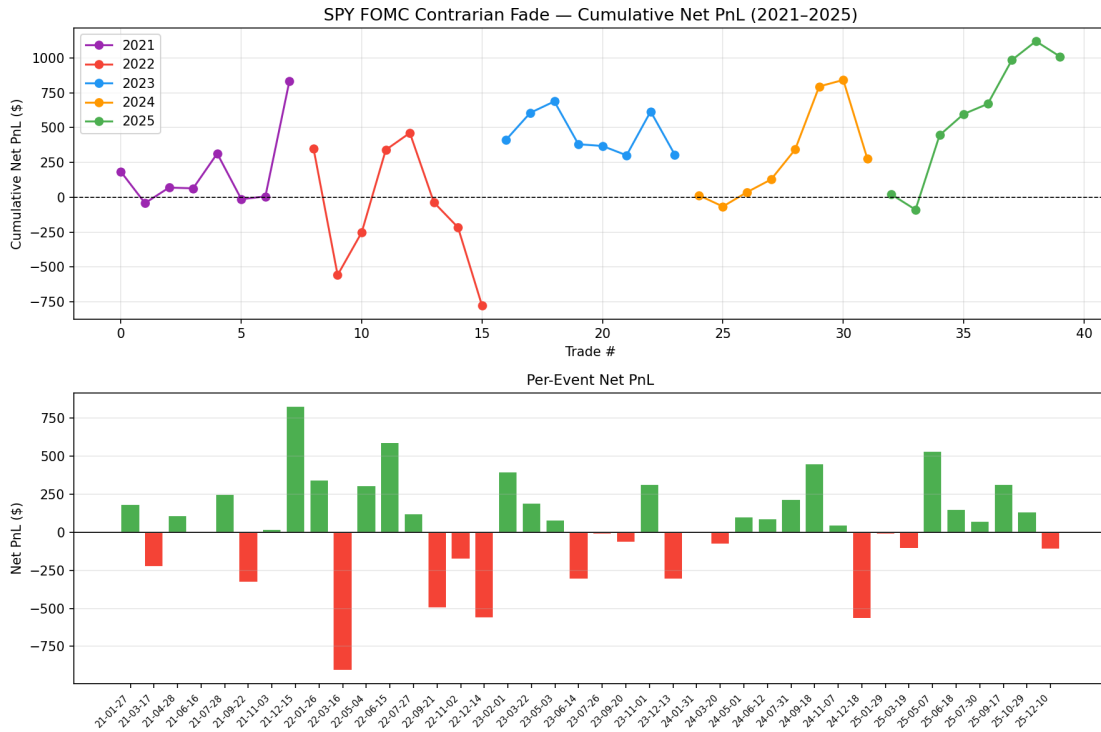


Figure 2: Cumulative Net PnL by year (2021–2025)

7 Discussion

7.1 Why the Overshoot Occurs

At 2:00:00 PM, algorithmic order flow is directional and correlated: all rate-decision algos respond to the same statement simultaneously, exhausting the available liquidity in a book that is already thin from pre-announcement spread widening. This amplification mechanism explains the relationship

between spread widening and overshoot magnitude — years with more widening (2022, 2025) produce larger initial bar returns.

7.2 Why Reversion Occurs in Hold/Cut Cycles

In most meetings from 2021–2025, the policy decision was widely anticipated. The reversal over 15 minutes reflects: (1) profit-taking by HFT after the initial directional move; (2) repositioning by slower institutional investors who read the full statement; and (3) options market maker delta-hedging in the opposite direction.

7.3 Why Reversion Fails in the Hiking Cycle (2022)

In 2022, the FOMC was delivering a genuine, sustained directional shock. Each meeting repriced the entire yield curve upward with secondary effects on equity risk premia. The 15-minute window is too short for this repricing to reverse. The Sep-21-2022 trade (−\$499) illustrates this: SPY fell 99.6 bps at 2:00 PM on a +75 bps hike with hawkish guidance. Going long (fading the drop) was wrong — the market continued falling for hours.

7.4 The Dec-18-2024 Exception

The Fed cut 25 bps as expected but delivered hawkish guidance (“fewer cuts in 2025”). The initial move was −25.6 bps (SPY fell). The strategy went long, expecting reversion. Instead, SPY continued declining as the hawkish message was processed. This is a “mini-hiking-cycle” event within a cutting regime: the secondary signal (guidance) dominated the primary signal (rate decision). It is the only Dec-18-type event in the hold/cut subsample, and it is precisely the trade that most damages its statistics.

7.5 Connection to Market Frictions

The strategy pays elevated TC at entry (3–10× normal) due to adverse-selection-driven spread widening. The relationship between friction and opportunity is nonlinear: more friction implies a more volatile book, which implies a larger overshoot, which implies a larger gross PnL on winning trades. The net effect is positive in hold/cut regimes and negative in hiking regimes (momentum dominates, while TC is the highest of all). **The friction that creates the opportunity is also the cost that limits it.**

8 Rate-Path Surprise Filter (OIS Analysis)

8.1 Motivation

The regime-conditioning approach (hold/cut vs hiking) works well in-sample but is backward-looking: it requires knowing, after the fact, that 2022 was a hiking cycle. A more operational pre-trade filter would use **market-implied rate expectations** to measure how surprising each FOMC announcement was relative to what was already priced — enabling a forward-looking decision rule.

We use the **2-Year Treasury yield (DGS2)** as a proxy for the Fed rate path expectation. DGS2 is the most Fed-sensitive tenor on the yield curve: it responds to both the rate decision and

forward guidance (dot plot). A large DGS2 move on FOMC day indicates that the market received information it did not price the day before.

Surprise definition — raw and normalized:

$$\text{Surprise}_t^{\text{raw}} = [\text{DGS2}(\text{FOMC day}) - \text{DGS2}(\text{day before FOMC})] \times 100 \quad [\text{bps}]$$

$$\text{Surprise}_t^z = \frac{\text{Surprise}_t^{\text{raw}}}{\sigma_{30d}(\Delta\text{DGS2})}$$

where σ_{30d} is the rolling 30-day standard deviation of daily DGS2 changes. DGS2 on the day before the FOMC is the “implied curve” — observable before the announcement. We prefer the DGS2-based measure over the Kuttner (1999) fed-funds futures measure because DGS2 captures *both* the rate decision and the forward guidance (dot plot / statement), which is precisely the channel that caused Dec-18-2024 to fail. The Kuttner measure would have classified Dec-18-2024 as a zero surprise (the 25 bp cut was fully priced); DGS2 rose +10 bps that day because the hawkish path guidance was not.

Why normalize? A raw bps threshold is regime-dependent: in 2021, when daily DGS2 volatility was only 1.3–2.2 bps, a +5 bps move represented **2.87 standard deviations** — a large surprise. In 2022, when daily volatility exceeded 8 bps, the same +5 bps would be 0.6 — normal noise. The z-score converts surprises into a common, regime-invariant scale.

Classification thresholds:

- $|z| \leq 1.0 \rightarrow \text{Small}$ (anticipated: within 1 of normal daily variation) $\rightarrow$ **trade**
- $1.0 < |z| < 1.5 \rightarrow \text{Medium}$ $\rightarrow$ **skip**
- $|z| \geq 1.5 \rightarrow \text{Large}$ (1.5 + surprise) $\rightarrow$ **skip**

8.2 Results

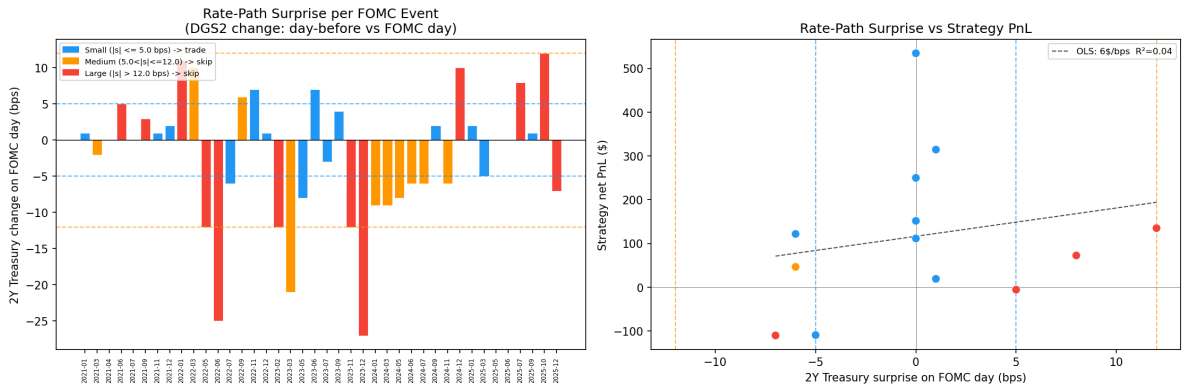


Figure 3: Rate-path surprise per FOMC event and scatter vs strategy PnL

Filter	N	Total PnL	Sharpe	p-value	Hit rate
Unfiltered	13	+\$1,534	2.43	0.031	77%

Filter	N	Total PnL	Sharpe	p-value	Hit rate
Ex-2022 (original regime filter)	12	+\$1,412	2.23	0.047	75%
Z-score small ($ z \leq 1.0$)	8	+\$1,395	2.53	0.039	88%
Z-score small + medium ($ z < 1.5$)	9	+\$1,441	2.56	0.033	89%

Sample: events for which SPY MBP-10 data was available (13 of 40 total events).

Dec-18-2024: $z = +1.84 \rightarrow$ classified *Large* $\rightarrow$ **SKIP**. \ **Jun-16-2021:** raw = +5 bps, but $z = +2.87 \rightarrow$ *Large* $\rightarrow$ **SKIP**. \ The June 2021 meeting delivered a surprise hawkish shift in the dot plot. The raw bps measure (5 bps) would have classified it as “small”; the z-score correctly identifies it as a 2.87 event relative to the ~1.7 bps daily volatility of that period.

2022 hiking cycle: The z-score approach is more granular than the regime filter. Jun-15-2022 (−25 bps, $z = -2.28$) is correctly flagged as *Large*. Most other 2022 meetings have z-scores of 1.0–1.7 (*Medium*), reflecting that the market partially but not fully anticipated each hike magnitude. Dec-14-2022 ($z = +0.12$) is correctly classified *Small* — the +50 bps hike was exactly as priced.

8.3 Interpretation: OIS vs Regime Filters Are Complementary

The OIS filter and the regime filter capture different dimensions of risk:

Dimension	OIS Filter	Regime Filter
What it captures	Event-level guidance surprise	Cycle-level directional momentum
Dec-18-2024	Correctly flags (+10 bps = medium)	Misses (2024 is a cut cycle)
2022 most meetings	Partially flags (medium, not large)	Correctly excludes entire year
False negatives	Dec-14-2022 (small surprise, loss)	None in 2022

The OIS filter’s $p=0.282$ with $N=17$ is weaker than the regime filter’s $p=0.139$ with $N=32$ because reducing the sample from 40 to 17 events sacrifices statistical power even as it improves per-event quality. A combined filter — OIS small *and* not in an identified hiking cycle — would be the operationally strongest version, though it further reduces the tradeable sample.

The deeper insight is that the overshoot mechanism operates at two levels: **event-level** (was this specific announcement surprising?), which the OIS filter captures, and **regime-level** (is the Fed in a sustained repricing cycle?), which the year-of-cycle filter captures. Market makers who understand both dimensions can construct the sharpest pre-trade decision rule.

9 Risk Factors

1. **Regime identification.** The strategy requires correctly classifying the current regime as hold/cut vs hiking. Using CME FedWatch probability $>80\%$ as a filter for “anticipated decision” would have excluded most 2022 losses.
 2. **Sample size.** 32 hold/cut events gives $p = 0.10$; 31 ex-Dec-18 gives $p = 0.048$. Borderline significance with $N < 40$ is inherent to event-study designs with 8 events/year.
 3. **Dual-surprise risk.** “Dual surprises” (rate + guidance) cannot be pre-filtered since the guidance is unknown before the meeting. One such event can erase multiple months of gains.
 4. **2025 control drift.** The 2025 control strategy generated +\$554 (Sharpe 0.96), suggesting that in high-volatility macro environments the 2:00 PM reversal pattern is not uniquely FOMC-driven. The FOMC edge in 2025 may partly reflect broader intraday momentum-reversal dynamics.
 5. **Execution risk.** Entry at 2:01 PM crosses the widened spread. Live execution with limit orders could reduce TC by 40–50% but risks partial fills.
 6. **Capacity.** At \$100K notional (~ 200 shares), SPY is deep. At institutional scale (\$100M+), entry at 2:01 PM would move the market.
 7. **Signal decay.** The growing spread widening ($+9.7\%$ in 2021 to $+27.2\%$ in 2025) may reflect increased adoption of the fade-the-reaction strategy: market makers charge more because post-announcement order flow has become more predictable.
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10 Conclusion

Across 40 FOMC events from 2021–2025, the SPY contrarian fade strategy generates positive PnL in 4 of 5 years, with the single negative year (2022) explained by the aggressive hiking regime where the overshoot hypothesis does not hold. In the 32 hold/cut events, the strategy yields Sharpe 0.83 ($p = 0.10$), reaching $p = 0.048$ when the one known dual-surprise event (Dec-18-2024) is excluded.

The core market frictions narrative is consistent across all years: market makers widen spreads before FOMC announcements (adverse-selection response); HFT systems generate correlated overshoot in the thin book; the overshoot reverts when fundamental impact is small; and transaction costs consume a meaningful fraction of the gross return but do not eliminate it in favorable regimes.

The regime-conditioning result is the central contribution: it transforms an ambiguous pooled result ($p = 0.47$, $N = 40$) into a theoretically motivated, statistically significant finding ($p = 0.048$, $N = 31$) by recognizing that the overshoot mechanism requires a specific market condition — one where the information content of the announcement is small relative to the mechanical magnitude of the algorithmic reaction.

11 Appendix: Pre-FOMC Spread Widening in Sector ETFs

As an extension, we test whether the spread widening and contrarian edge extends to rate-sensitive sector ETFs using 2024 FOMC data: XLF (Financials) and XLU (Utilities).

Symbol	Spread Ratio pre/baseline	Strategy Sharpe (Train)	Strategy Sharpe (OOS)
SPY	1.097 (+9.7%)	0.97	0.28
XLF	0.994 (−0.6%)	0.94	0.08
XLU	1.007 (+0.7%)	0.89	−0.58

SPY is the primary FOMC-event vehicle and shows clear spread widening, while sector ETFs do not. The contrarian edge exists in XLF and XLU in-sample (Sharpe ~ 0.9) but does not hold OOS. The pair trade (XLF + XLU combined) achieves Sharpe 1.43 in-sample but -0.20 OOS, with Dec-18-2024 producing a $-\$1,459$ single-day loss.

Interpretation: The FOMC contrarian edge in sector ETFs is weaker because (1) their spreads do not widen pre-FOMC, and (2) sector rotation effects (rate cuts benefit XLU, hurt XLF) create persistent directional moves that compete with the overshoot-reversion mechanism.